BHARAT SIRMAL

sirmalbharat99@gmail.com | LinkedIn | GitHub | Portfolio | Mumbai, Maharashtra, India

EDUCATION

B.Tech in Computer Science *2023 - 2027*
Pillai College of Engineering, Panvel
CGPA: 8.74 / 10.0

Class 12 (Higher Secondary) *2020 - 2022*
KMC Balkumari, Lalitpur Nepal
GPA: 3.14 / 4.0

SKILLS

Languages: Python, JavaScript, C++, SQL
AI / Machine Learning: PyTorch, CNNs, RAG, LLMs
Backend / Frontend: Flask, FastAPI, Node.js, Express.js, REST APIs, React, Next.js, HTML, CSS, Tailwind CSS
Databases / Tools: MongoDB, MySQL, SQLite, ChromaDB, Firebase, Git, GitHub, Postman, Google Cloud, Vercel, Cloudinary

PROJECTS

Explainable Brain Tumor Classification
- Developed a 2-stage multimodal AI pipeline that screens 3 MRI views for tumor presence and classifies positive screens into 4 tumor types, incorporating 2 clinical inputs-age and sex.
- Integrated 2 explainability methods, Grad-CAM and Shapley attribution, with scan-specific interpretations.
- Built a responsive dashboard with input verification, result-specific guidance, and automated PDF reporting.
- Technologies: PyTorch , CNN, MRI/NIfTI, Grad-CAM

Bharat AI - Personal AI Assistant (RAG)
- Developed an end-to-end Retrieval-Augmented Generation (RAG) assistant using a private knowledge base, semantic retrieval, and Gemini for grounded responses.
- Built an embedding pipeline with all-MiniLM-L6-v2 and ChromaDB, including similarity-based retrieval and relevance filtering.
- Implemented role-based data access control, enforcing strict segregation between admin and user records with zero unauthorized access.
- Integrated the AI backend with a conversational Next.js interface supporting streaming responses and voice interaction.
- Technologies: Python , Flask , Gemini , ChromaDB, Next.js

JoJo Store - Full-Stack E-Commerce Platform
- Developed a responsive e-commerce platform with product browsing, shopping cart, secure authentication, and Stripe payment integration.
- Built an admin dashboard for managing products, inventory, customers, orders, and delivery assignments.
- Integrated Firebase for authentication and data storage, Cloudinary for image uploads, and role-based access for customers, administrators, and delivery partners.
- Implemented role-based access for 3 user roles and validated core e-commerce workflows with 64 automated tests.
- Technologies: Next.js , TypeScript , Firebase , Stripe

YAOP Community Platform
- Designed and managed a centralized MongoDB database, organizing structured data for events and volunteers with full accuracy and consistency across all records.
- Processed and validated donation entries with complete data integrity by implementing Firebase authentication and structured data workflows.
- Built an admin dashboard that streamlined data entry and record updates, reducing manual effort by ~40% and minimizing input errors.
- Technologies: Next.js, Express.js, MongoDB, Firebase

INTERNSHIP

Web Development Internship *May 2026 - Jun 2026*
Company Name: InAmigos Foundation
Assisted in developing and maintaining the organization's website using HTML, CSS, and JavaScript. Designed and implemented new features to enhance site functionality, and improved the layout and structure to make the website more visually attractive and easier to manage.
Technology Used: HTML, CSS, JavaScript, React, Node.js, MongoDB

CERTIFICATIONS & ACHIEVEMENTS

Google Cloud Data Analytics Certificate
Hands-on experience with cloud-based data pipelines, data analysis, and structured reporting tools.

Google Cloud Career Launchpad (Foundations Track)
Training in cloud computing fundamentals, data management, and analytics best practices.